CNC Procedure 2018_001

Jan 9th

2018

Information on Touch Panel Replacement and Calibration in FANUC Open Systems

[Applicable for all Fanuc Open CNC with EWF function (Component of Windows)]

How to replace and calibrate Touch Panel in Open CNC

(1) Confirmation of touch panel response

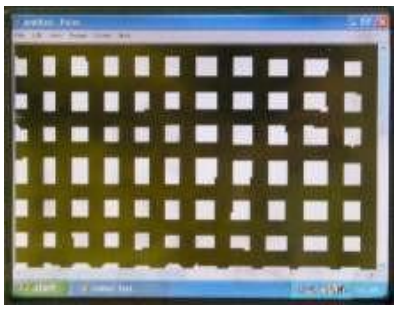
※When you operate the touch panel, please use the touch panel pen.
If you use something like a ballpoint pen, you may damage the touch panel.

Confirm the response of touch panel by the following method, and then evaluate the target for replacement activity:

1 Activate "Paint" which is an application of Windows OS.
（Start Menu → Program → Accessory →"Paint"）

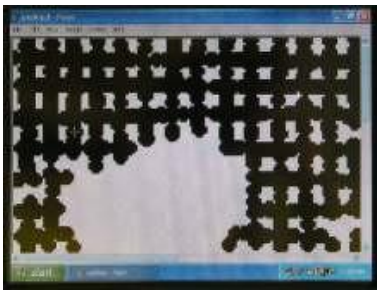
- Draw lattice pattern that shown in the figures below.
- How to evaluate (Example of drawing lattice pattern in all areas）

Case A: Lattice pattern can be drawn in all areas.



⇒Normal case

Case B: Part of lattice pattern cannot be drawn.
Lattice pattern can only be drawn in one part of area.



⇒ Abnormal case

Case C: (Example of not targeted failure) Lattice pattern cannot be drawn in all areas.
Case D: (Example of not targeted failure) When you cannot draw lattice pattern in all areas.

(2) Life expectancy of touch panel under normal working condition is 5+ years.

(For Example only)
Preparation of touch panel parts for replacement

(1) Touch panel unit A08B-0084-D263

(2) Touch panel interface board

The basic unit is A13B-0196-B123⇒ A20B-8001-0621

The basic unit is A13B-0196-B031⇒ A15L-0001-0086#A (Vender model number：N16B-0558-0540)

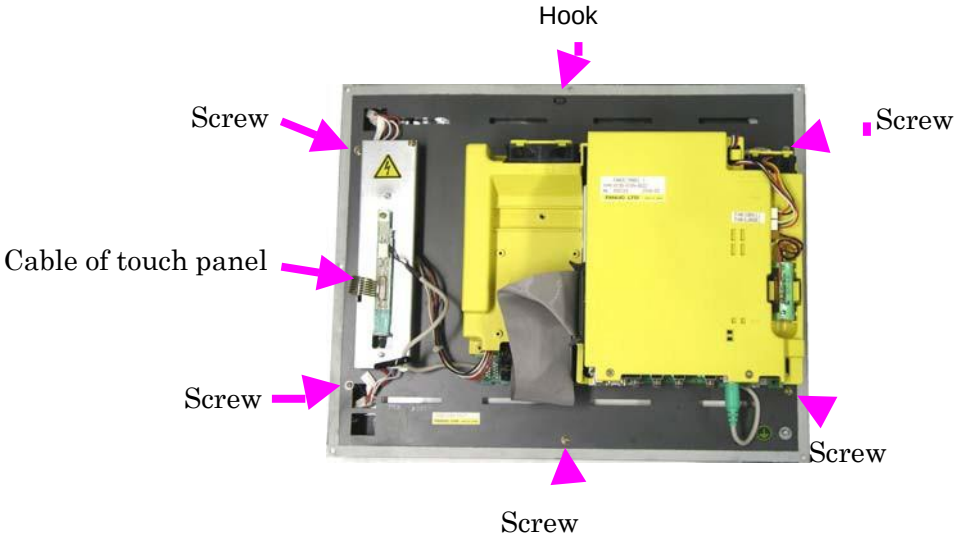
(3) Touch Panel Pen

2. Method of replacement

(1) Remove unit from cabinet.

(2) Remove the touch panel cable and the screws from five different points.

※ In order to remove the touch panel cable, you need to remove the latch of the connector. Be careful to pull out the cable. If you put too much strength, you may damage the cable.



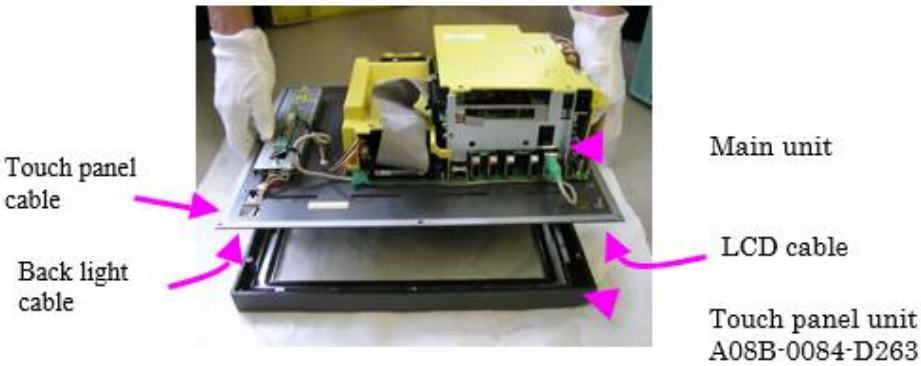
(3) Prepare a new touch panel unit.（A08B-0084-D263）

※ Clean up dusts inside the touch panel with air spray.

(4) Remove the touch panel unit from the main unit while removing the hook of cover.

※ The touch panel is stuck to sheet metal by packing. Remove the touch panel from bottom.

※ Please be careful not to allow the touch panel cable get stuck inside the hole of sheet metal.



(5) Install the touch panel unit to main unit while fitting in holes of screw.

※ Please be careful to avoid dusts inside the touch panel and the surface of LCD panel.

※ When installing the touch panel, please, be careful not to put the touch panel cable, the LCD cable and the back light cable between the sheet metal and the touch panel unit. Since there is a possibility of pinching the LCD cable at the insertion port of PCMCIA card, it is necessary to handle this operation very carefully.

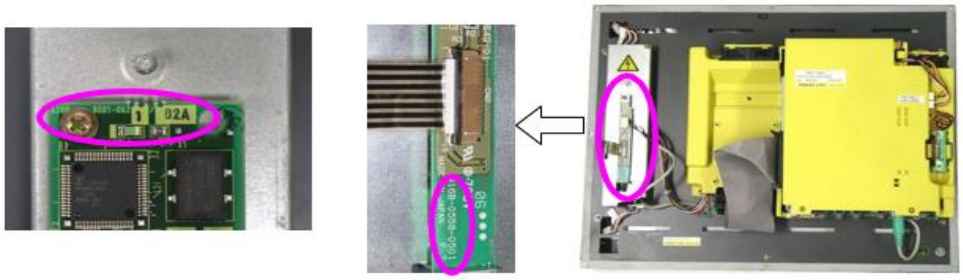
(6) Screw the five points.

(7) Confirm that dusts are not between the panel and LCD panel.

(8) If touch panel interface board is old type, replace the old board with a new one.

(9) Touch Panel interface board for reference

Panel-i Storage	CF card type (A13B-0196-B123)	HDD type (A13B-0196-B031)
Old type	A20B-8001-0620	N16B-0558-0501
New type	A20B-8001-0621	N16B-0558-0540



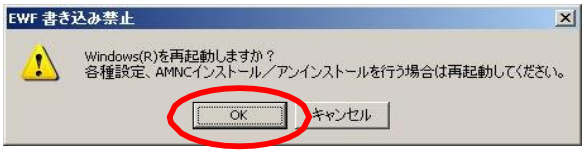
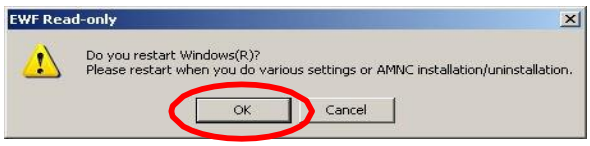
(10) Plug in the touch panel cable to connector and close the latch.

(11) Mount the unit to the cabinet.

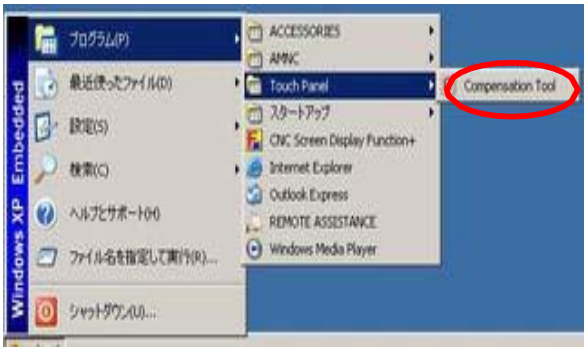
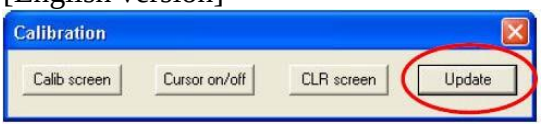
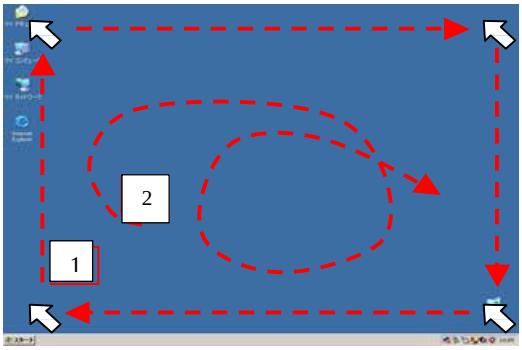
(12) Turn on the PANEL i, then make sure that you can operate the PANEL i properly

(1) Confirm that the PANEL i can start properly
(2) Confirm that there are not faults on the screen (color deviation, misalignment, blur).
(3) Confirm that touch panel can be drawn in all areas (Use Step 1~1-3)

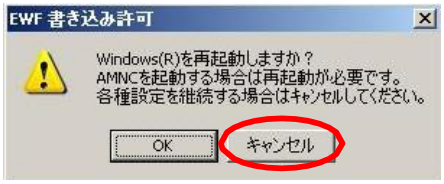
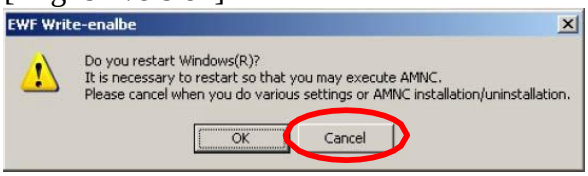
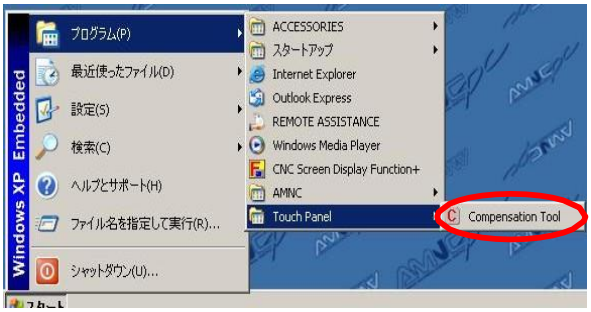
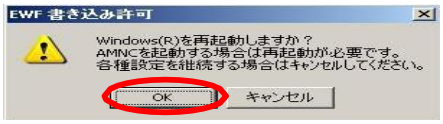
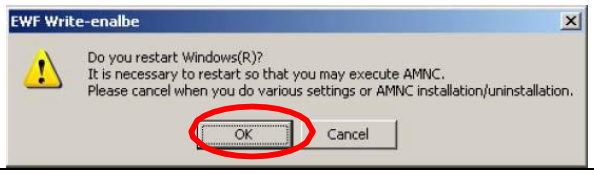
(13) Calibrate the touch panel referring the below procedure:

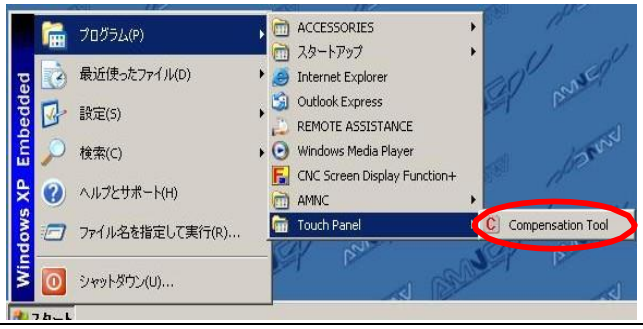
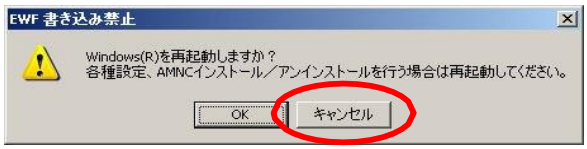
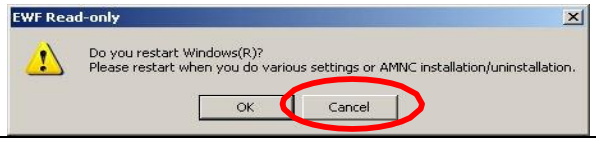
Touch Panel Readjustment manual（CF Card Type)	
Procedures	**Screen**
Turn on the PANEL i and start up Windows. 【Note】 MTB User screen is displayed, but an error message for NC bad connection will be displayed without activating the HMI application in case that the HSSB optical cable or Ethernet Cable is not connected to CNC incase if standalone.	
Activate the "Compensation Tool" of Touch panel	
The message ("EWF read-only") will appear on the screen. Then, click on the button that say "OK". After that, Windows will reboot automatically.	[Japanese version]  [English version] 
Login screen will display. Then, type USER NAME **(Administrator) and** PASSWORD **(OPENCNC).** Finally, click on the button that say "OK" 【Note】 Push of the TAB key, when you move the cursor. Keep on pushing the SHIFT key	

Touch Panel Readjustment manual（CF Card Type)

Procedures	Screen
The Windows OS will be activated as an administrator. （Wallpaper is not displayed.） 【Note】 EWF has been deactivated at this point.	
Activate the "Compensation Tool" of Touch panel	
Calibrate the touch panel by touching red points Red cross logos will move after touching each points. Touch red cross logos in all points. After touching all points, screen will move to next screen automatically.	
Calibration will be completed by clicking on the button that say "Update".	[Japanese version] [English version] 
Confirm the position shifts of cursor in each corner of screen and all area by moving the cursor like ① and ② Also, confirm no misalignment and following up a cursor.	

Touch Panel Readjustment manual（CF Card Type）

Procedures	Screens
The message of "EWF write-enable" will be displayed after completing adjustments. Then click on the button that say "Cancel" (Attention） Do not click on the button that say "OK" (If you click on the button that say "OK", windows will be rebooted. Then EWF will be activated.) 【Note】 If there are misalignment of cursor, conduct the calibration settings again (items No,7-11).	[Japanese version]  [English version] 
Select "shutdown" and click on the button that say "OK" at shutdown screen After shutdown the windows, turn off the PANEL i once	
Turn on the PANEL i and start up Windows again. The same screen with the first item will be appeared. 【Note】 In this point, EWF has been deactivate continuously.	
Activate the "Compensation Tool" of Touch panel	
Calibrate the touch panel in a similar manner. (Please refer to items of No,8-10). After completing adjustments, the message of "EWF Write-enable" will be appeared. Click on the button that say "OK" (Windows will be rebooted automatically.) 【Note】 In case that there are misalignment of cursor, click on the button that say "Cancel" Then, conduct the settings again (items No14～15)	[Japanese version]  [English version] 

Touch Panel Readjustment manual（CF Card Type）	
Procedures	Screens
Windows will launch again. 【Note】 In this point, EWF has been activated.	
Confirmation of the state of AMNCPC which is factory default settings (EWF has been activated).】 Activate the "Compensation Tool" of Touch panel	
If the message of "EWF Read-only" will be displayed as shown in right figure, EWF is activated. (factory default settings) Then click on the button that say "Cancel" （Attention） Do not click on the button that say "OK" （If you click on the button that say "OK", windows will be rebooted.) 【Note】 In case that the touch panel adjustment screen (item of No, 8) is displayed, EWF is deactivated. (In this case, the right figure is not displayed) Conduct settings again(item of No,14)	[Japanese version]  [English version] 
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